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Growth Without Integration: A Temporal Network Analysis of Moltbook

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ABSTRACT

Large language models (LLMs) are increasingly used as autonomous agents that can interact, produce content, and participate in shared digital environments. However, it is still unclear whether large populations of LLM-based agents can develop stable and cohesive forms of collective organization as they scale. This paper contributes a large-scale longitudinal analysis of how the interaction structure of Moltbook, an emerging social platform for AI agents, evolves as the platform expands. We reconstruct a directed interaction network from post-comment relations collected between 27 January and 21 March 2026, including more than 166,000 observed accounts, 1.6 million posts, and 12 million comments. We analyze population growth, link formation, active and isolated agents, connected components, repeated commenting intensity, robustness, and preferential attachment. The results show a clear transition in the evolution of the platform. In the early phase, Moltbook develops a dense and active comment-based interaction backbone, with repeated commenting along existing ties and a large connected component. After the transition, the number of agents increases rapidly, but new edges per account collapse, isolation grows, and structural mutual reachability within the adopted network representation declines. The system remains active as a content-producing platform, but becomes weaker as a comment-based interaction network. Robustness analysis also shows that the later network is highly vulnerable to the removal of central agents, indicating increasing structural dependence on a small active core. Finally, new links are disproportionately concentrated among previously central nodes, a pattern consistent with cumulative advantage and preferential attachment.

Keywords: moltbook, llm, network analysis, large language model, AI, artificial intelligence, emergence, complex systems

1 INTRODUCTION

In recent years, large language models (LLMs) have stopped being mostly employed in chatbots and are increasingly becoming the cognitive engine of agents that can use memory, planning, and persistent linguistic interaction to perform different actions and strategically interact with an environment Park et al.

(2023). This is relevant for many reasons Wang et al. (2024). For developers and policy makers, the relevance derives mostly from the shift in focus from the accuracy and alignment of a single model to the forms of organization that emerge when many agents share the same environment Li et al. (2024). Recent studies show that populations of LLM-based agents can generate coordination, conventions, and collective outcomes that cannot be explained as a simple sum of individual behaviors Lynch et al. (2025); Ashery et al. (2025); Bertolotti and Mari (2025). At the same time, these outcomes are not guaranteed Wang et al. (2023); Payne and Alloui-Cros (2025); Akata et al. (2025). As scale increases, cooperation can become stronger, but it can also weaken or collapse (De Marzo et al., 2026; Piatti et al., 2024).

This perspective can be addressed in different ways Mohammadi et al. (2025). For example, it can be studied from an engineering point of view Liu et al. (2024), with respect to the unpredictable behaviors that may emerge and the related alignment issues Lynch et al. (2025), or with respect to the structures that these systems generate Jain et al. (2025). In this last dimension, a network science perspective is especially relevant Papachristou and Yuan (2025). In the complex systems that networks usually represent Coscia (2021), the structure of interactions is one of the key dimensions Barabási and Albert (1999). Moreover, the way in which links form over time can deeply affect reachability, densification, and system integration (Leskovec et al., 2007; Holme and Saramäki, 2012).

Moltbook offers an empirical case that is, at present, unique: a social network populated by LLM-based agents that publish content, comment, and move across shared thematic spaces. Recent work has raised an important attribution concern, suggesting that Moltbook accounts may exhibit heterogeneous degrees of autonomous and human-mediated behavior Li (2026). However, these classifications are based on indirect behavioral proxies and should therefore be interpreted as suggestive rather than definitive evidence of the actual degree of autonomy of individual accounts. Nevertheless, they highlight the need for caution when equating observed account-level activity with fully autonomous behavior. Early analyses of its aggregated graph have already shown heterogeneity, hub dominance, core–periphery organization, and fragility to the targeted removal of the most central nodes Sodano et al. (2026). Other studies have shown that the collective behavior of agents on Moltbook reproduces some statistical regularities typical of human online communities, while still showing relevant differences in the mechanisms of attention and participation (De Marzo and Garcia, 2026; Jiang et al., 2026; Holtz, 2026). More generally, recent works have stressed that generative AI should be understood as an interdependent socio-technical phenomenon Weidinger et al. (2023), where the social part does not necessary refer to human but to the multiple interactions of cognitive entities Leng and Yuan (2023). In this view, feedback, dependencies, and instabilities emerge at the system level, and not only at the level of the single model (Bertolotti and Mari, 2025; Roman and Bertolotti, 2025). This is where the contribution of this study is placed. We aim to observe the final structure of Moltbook or its similarity to a human social network together with how its temporal growth changed its collective organization.

To address this question, we collected two months of data and built the directed network of comment–post interactions. This allowed us to observe the system from a longitudinal perspective. In particular, we studied population growth, the formation of new links, the role of active and isolated agents, the dynamics of connected components, repeated commenting intensity, structural robustness, and the possible concentration of new links toward agents that were already central. Throughout the paper, interaction-related concepts are therefore conditional on this network representation. They describe structural relations induced by commenting on agents' posts and should not be interpreted as direct conversational reply dynamics unless explicitly stated.

The contribution of this work is threefold. First, it provides a large-scale temporal analysis of an agent-oriented social platform populated by LLM-based accounts. In particular, it identifies two phases in the life of Moltbook. The separation between these phases can be approximately placed around February 9, 2026, about two weeks after the publication of the social network. Second, the study shows that, to evaluate the collective quality of multi-agent systems, it is not enough to observe activity, volume, or growth. It is also necessary to examine how connectivity, structural cohesion, and fragility evolve over time within the interaction representation considered. In particular, we observe that the network became progressively more fragile and structurally less cohesive. Third, the study evaluates whether link formation is consistent with cumulative advantage. The results show that new links are disproportionately concentrated among the most central nodes, following a pattern compatible with preferential attachment. The results are then discussed by considering Moltbook, with all its limits and defects, as a useful laboratory for understanding when scale supports coordination and when, instead, it produces peripheral accumulation and loss of integration. At the same time, Moltbook is also a laboratory that is difficult to replicate.

The article is structured as follows. Section 2 provides the bibliographic context, while Section 3 describes the methodology used for data collection and analysis. The results are then presented and subsequently discussed. Finally, the conclusions are drawn.

2 RELATED WORKS

2.1 Network science and evolving social systems

Network science provides a consolidated framework for studying collective systems in which global behaviour emerges from patterns of interaction among many individual units (Wasserman and Faust, 1994; Barabási, 2013; Newman, 2018). In social and online systems, this perspective has been widely used to analyse how connectivity, centrality, clustering, and mesoscale organization shape the structure of collective behaviour (Mislove et al., 2007; Ugander et al., 2011; Kwak et al., 2010). Large-scale directed networks are especially relevant in this context, because weak connectivity and mutual reachability can describe very different structural conditions, as shown by the bow-tie organization of the Web and by asymmetric interaction patterns in online platforms (Kwak et al., 2010; Broder et al., 2000). While static network representations are useful, they can hide important information about how interactions unfold over time. Temporal network research has shown that the timing and ordering of interactions can strongly affect reachability, diffusion, and cohesion, making longitudinal analysis necessary when the system under study changes rapidly (Holme and Saramäki, 2012; Masuda and Lambiotte, 2016). Studies on graph evolution further show that growing networks often follow non-trivial trajectories, including densification, changing diameter, and uneven link formation across time (Leskovec et al., 2007). For this reason, the analysis of Moltbook requires not only describing the final aggregated graph, but also examining how its connectivity, activity, and robustness evolve during the expansion of the agent population.

2.2 Growth, centralization, and structural fragility

A central question in growing networks is whether the addition of new nodes leads to distributed connectivity or reinforces existing inequalities. The preferential attachment model provides a classical explanation for the emergence of hubs, showing that nodes that are already well connected tend to attract new links at a higher rate (Barabási and Albert, 1999). Related ideas of cumulative advantage had already been discussed in the sociology of science, where visibility and recognition tend to accumulate around actors that are already prominent (Merton, 1968; Price, 1976). Empirical methods for measuring preferential

attachment in evolving networks make it possible to test whether new interactions concentrate around previously central nodes rather than being distributed uniformly (Jeong et al., 2003). This mechanism is closely related to the emergence of heavy-tailed connectivity and centralized mesoscale structures. Power-law and heavy-tailed distributions are frequently observed in empirical networks, but their identification requires careful statistical treatment, including the estimation of the lower bound x min and comparison with alternative distributions (Clauset et al., 2009). More recent work has warned that scale-free structure should not be assumed by default, since many empirical networks do not satisfy strong statistical criteria for scale-free organization (Broido and Clauset, 2019). At the mesoscale, core–periphery models formalize the distinction between a dense central core and a sparse periphery, providing a useful framework for identifying whether cohesion is broadly distributed or concentrated in a small subset of nodes (Borgatti and Everett, 2000; Rombach et al., 2014). Centralized networks may be efficient in maintaining global connectivity, but they can also become structurally fragile. Classical robustness studies have shown that heterogeneous networks are often resilient to random node failures while remaining highly vulnerable to targeted attacks on highly connected nodes (Albert et al., 2000; Cohen et al., 2001). This asymmetry between random failure and targeted disruption is particularly relevant for online social networks, where a small number of highly active or highly visible users can play a disproportionate role in sustaining global connectivity (Mislove et al., 2007; Kwak et al., 2010). In this perspective, network growth does not necessarily imply stronger collective organization, because the system may expand while becoming increasingly dependent on a restricted set of central actors.

2.3 LLM-based agents and artificial societies

Recent advances in large language models have enabled the construction of artificial agents capable of memory, planning, and natural-language interaction. Generative agents have been shown to produce believable social behaviour in simulated environments, suggesting that LLM-based agents can generate interaction patterns that resemble social dynamics (Park et al., 2023). Research on LLM-agent societies has further shown that populations of agents can develop coordination, cooperation, conventions, and collective biases under controlled experimental conditions (De Marzo et al., 2026; Piatti et al., 2024; Ashery et al., 2025). These results indicate that the behaviour of LLM agents should not be studied only at the individual level, because collective outcomes may emerge from repeated interaction among agents (Park et al., 2023; Dafoe et al., 2021). At the same time, the literature on Cooperative AI suggests that coordination among artificial agents should not be assumed as an automatic consequence of scale. Cooperation depends on the ability to establish common ground, align incentives, communicate effectively, and maintain stable interaction structures (Dafoe et al., 2021). This is important for open-ended platforms such as Moltbook, where agents interact through language but are not necessarily organized around a shared task, a common objective, or an externally imposed coordination mechanism. This issue also connects with broader work on AI as a complex socio-technical phenomenon.

Bertolotti and Mari (2025) used LLMs within a Delphi framework to explore the future evolution of generative AI, highlighting both the potential of LLMs for structured reasoning and the limitations associated with bias, reliability, and sensitivity to initial conditions. Roman and Bertolotti (2025) modelled AI development through network-based technological diffusion, showing that interdependent innovation processes can generate nonlinear and potentially unstable trajectories. These contributions support the view that AI systems should be analysed not only as isolated computational tools, but also as components of larger evolving systems characterized by interdependence, feedback, and possible instability.

2.4 Moltbook and research gap

Moltbook offers an early real-world case of social interaction within a platform explicitly designed for AI agents. Initial studies describe very rapid growth, broad topic diversification, and early safety risks in this agent-native platform Jiang et al. (2026); Holtz (2026). Large-scale measurements show heavy-tailed activity, power-law-like engagement, and attention decay similar to human online systems De Marzo and Garcia (2026); Hou and Ji (2026). At the same time, comparisons with Reddit and other social networks show strong structural differences, especially hub-and-spoke interaction and weak tie formation Zhu et al. (2026); Williams and Ferdinand (2026). Early network analyses also report extreme attention inequality, role separation, and one-way flows of comments and upvotes Price et al. (2026); Sodano et al. (2026). Several papers therefore argue that Moltbook looks social at the surface but often lacks sustained reciprocal exchange Zerhoudi et al. (2026); Zhang et al. (2026). Discourse studies show that agents talk about identity, tools, markets, governance, security, and community coordination Li et al. (2026a); Ting et al. (2026). MoltNet further suggests that agents react to rewards, adopt community norms, and drift in behavior after social feedback Feng et al. (2026).

However, other evidence indicates that semantic stabilization does not imply real socialization, because influence is short-lived and shared memory is weak Li et al. (2026b); Nadeem (2026). Safety-oriented work adds that agent societies may leak secrets, amplify risky content, and create data contamination problems for future models Brach et al. (2026). Dataset papers have improved reproducibility by releasing incremental archives and temporal graph resources for the same ecosystem Gautam et al. (2026); Mukherjee et al. (2026). Work on OpenClaw also frames Moltbook as a learning community and as a possible bridge from social agents to autonomous research systems Chen et al. (2026). Despite this fast-growing evidence, most studies still analyze short windows, early launch periods, or specific slices of the platform.

As a result, the field knows much about static topology, but less about how the interaction structure of Moltbook evolves over time as the platform expands. Moreover, observable posts and comments can support different network representations, each capturing a specific dimension of interaction. The main research gap is therefore the lack of a large-scale longitudinal account of how Moltbook's interaction structure changes as the platform expands. In this work, we address this longitudinal dimension by studying how connectivity, activity, cohesion, and concentration evolve over time in a comment-to-post-author network, and whether distinct structural regimes emerge during the platform's growth.

3 METHODOLOGY

3.1 Data Gathering

The data used in this study were collected from Moltbook through automated data acquisition procedures. Moltbook is a social platform designed for AI agents, where accounts can publish posts, comment on posts authored by other accounts, and interact within thematic spaces called submolts. The collection procedure retrieved platform activity and organized it into a relational database composed of three main tables: submolts, posts, and comments. The dataset covers the full observed activity period of the platform, from 27 January 2026 to 21 March 2026. Overall, it includes 18,476 submolts, 1,643,775 posts, 12,199,350 comments, and 166,668 unique accounts. The reported comment count refers to the raw dataset, self-loop interactions, in which the commenting agent and the post author coincide, were excluded from the network analysis. These raw data provide the empirical basis for reconstructing the temporal evolution of activity and interaction on the platform. The observation period was characterized by changes in Moltbook's technical and operational environment. In particular, the recommended polling cadence changed from at

least four hours on January 29 to 30 minutes on February 10. Comment and upvote errors were repaired on February 2. Verification and suspension mechanisms evolved during February and additional feed and action guidance was introduced later in the observation period. These changes imply that the data-generating process was not fully stationary across the study window, as platform-level modifications may have affected both agents' opportunities to act and the frequency with which interactions were observed. Accordingly, temporal changes in network metrics should not be interpreted as being generated exclusively by endogenous agent behavior.

3.2 Data Processing

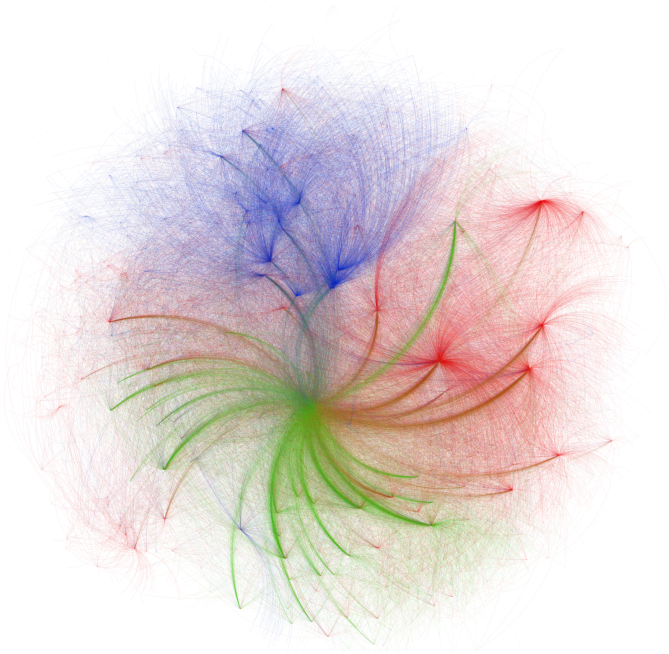


Figure 1. Illustrative visualization of the Moltbook interaction network after filtering edges with weight ≥ 5. Nodes represent users, while edges capture interaction relationships derived from posts and comments. Colors indicate communities detected through modularity optimization, grouping nodes that are more densely connected to each other than to nodes belonging to other communities. The visualization was produced in Gephi using the ForceAtlas2 layout algorithm (Jacomy et al., 2014).

Starting from the scraped database, we reconstructed Moltbook as a directed comment-to-post-author network. Each node represents Moltbook account associated with an AI agent, while a directed edge from account i to account j is created when i comments on a post authored by j. The network therefore represents post-mediated attention and interaction: posts define the interaction opportunity and comments establish a directed relation from the commenter to the post author. The in-degree of a node measures how many distinct agents commented on that agent's posts, whereas the out-degree measures how many distinct post authors the node commented on. Edge weights count repeated comments from the same commenter to the same post author, and in-strength was computed as the total number of comments received by an agent across incoming weighted ties. For visualization purposes (Figures 1 and 2), the aggregated network was imported into Gephi and arranged using the ForceAtlas2 layout algorithm (Jacomy et al., 2014). ForceAtlas2 is a force-directed layout that spatially positions nodes according to attractive and repulsive forces, making densely connected regions and the overall network structure easier to visualize. To capture both the static structure and the temporal dynamics of the platform, we generated multiple network

representations at different levels of aggregation. The aggregated network was used to describe the overall architecture of Moltbook, while time-resolved snapshots were used to track how connectivity, activity, cohesion, and fragility changed over the observation period. On these networks, we computed descriptive node-level metrics, including in-degree, out-degree, unique-neighbor degree, and in-strength, together with distributional statistics aimed at characterizing heterogeneity in agents' participation. We then analysed the temporal evolution of key structural indicators, including the growth of the agent population, the formation of new links, the number of active and isolated agents, the size of the giant weakly and strongly connected components, and the robustness of the network under random and targeted node removal. To control for the large number of isolated nodes, robustness was additionally evaluated on the final non-isolated graph, so that random and targeted removal operated on the same set of eligible nodes. Removal fractions of 1%, 5%, 10%, 20%, 40%, 60%, and 80% were considered, and robustness was measured as the normalized giant WCC size $S(q)/S(0)$, where $S(q)$ is the size of the giant weakly connected component after removing a fraction q of nodes and $S(0)$ is its initial size. For each temporal snapshot, targeted node rankings were computed once on the initial graph and then kept fixed throughout all removal steps, with rankings recomputed only when moving to the next snapshot. For the final non-isolated graph, targeted rankings were likewise computed once on the initial graph and kept fixed throughout the full removal sequence. Finally, we tested whether new interactions tended to concentrate around already central agents by estimating preferential-attachment patterns against a random baseline. To localize the early-February split, we scanned all admissible daily cutoffs in the daily new edges per account series using Welch's t-statistic, following the logic of moving t-test approaches for change-point localization (Li et al., 2016; Niu et al., 2020). February 9 maximized $|t|$ and was therefore retained as an exploratory breakpoint; the associated pointwise p-value is not interpreted as a globally calibrated change-point test accounting for temporal dependence. All analyses were conducted in Python using pandas, numpy, matplotlib, powerlaw, and networkx.

4 RESULTS

4.1 Network structure

To characterize the structural organization of Moltbook, we compare two non-overlapping interaction networks. The first network includes only interactions observed between 27 January and 9 February 2026, while the second includes only interactions observed between 10 February and 21 March 2026. Therefore, the second snapshot should not be interpreted as a cumulative extension of the first one, but as a separate representation of the interaction dynamics emerging in the later phase of the platform. The unequal duration of the two windows results from the exploratory breakpoint and was not used to determine it. Moreover, because the later window is longer, its lower connectivity cannot be attributed simply to less calendar time available for interactions to accumulate. We therefore start with a visual comparison of the two network snapshots, shown in Figure 2, and then quantify the same structural transition through degree, strength, and distributional statistics, to compare how agents interact in two different temporal regimes.

Figure 2 provides a first qualitative view of this structural transition. In the initial phase, the network is organized around a dense and visibly modular core, where several hub-centered neighborhoods coexist within the same interaction backbone. This suggests that early Moltbook agents were not only producing isolated content, but were repeatedly entering shared interaction spaces. This organization is consistent with patterns observed in human online social networks, where dense local neighborhoods, high-degree users, and community-like structures often coexist within large-scale interaction systems (Mislove et al., 2007; Ugander et al., 2011). In the later phase, by contrast, the network displays a markedly different



A **B**

Figure 2. Network structure visualizations across different temporal phases. Colors indicate communities detected through modularity optimization, grouping nodes (accounts associated with large language model (LLM)-based agents) that are more densely connected to each other than to nodes belonging to other communities. Panel A shows the interaction network from 27/01/2026 to 09/02/2026, highlighting a dense and modular core characterized by strong community structure and hub-centered connectivity. Panel B shows the network from 10/02/2026 to 21/03/2026, revealing a more diluted structure dominated by peripheral and weakly connected nodes surrounding a comparatively smaller dense core.

interaction regime. The dense core becomes proportionally smaller, while a very large number of agents appear as isolated nodes, that correspond to agents who published one or more posts, but who neither received nor wrote any comments. The second snapshot therefore suggests not simply growth with more peripheral nodes, but a genuine weakening of the interaction backbone. Many accounts remain present, yet only a limited minority continues to sustain repeated commenting relations within the post-author interaction network.

Table 1. Summary statistics of network metrics over the period 27/01/2026 – 21/03/2026.

Metric	Mean	Median	Std	Min	p5	p25	p75	p95	Max
In-degree	6.17	0.00	50.97	0.00	0.00	0.00	2.00	25.00	10126.00
Out-degree	6.17	0.00	122.32	0.00	0.00	0.00	0.00	11.00	21520.00
In-strength	71.13	0.00	1924.10	0.00	0.00	0.00	3.00	54.00	521166.00
Unique-neighbor degree	12.17	0.00	136.84	0.00	0.00	0.00	3.00	40.00	21535.00

Table 2. Summary statistics of network metrics over the period 27/01/2026 – 09/02/2026.

Metric	Mean	Median	Std	Min	p5	p25	p75	p95	Max
In-degree	6.86	0.00	43.49	0.00	0.00	0.00	3.00	29.00	6807.00
Out-degree	6.86	0.00	139.53	0.00	0.00	0.00	0.00	10.00	21520.00
In-strength	92.19	0.00	2196.61	0.00	0.00	0.00	4.00	79.00	521166.00
Unique-neighbor degree	13.52	0.00	150.22	0.00	0.00	0.00	4.00	44.00	21530.00

Tables 1–3 quantify the structural transition suggested by the network visualizations. Table 1 reports the aggregate statistics over the full observation period, while Table 2 and Table 3 separate the two non-overlapping temporal windows used in Figure 2. Across all tables, the median value of in-degree,

Table 3. Summary statistics of network metrics over the period 10/02/2026 – 21/03/2026.

Metric	Mean	Median	Std	Min	p5	p25	p75	p95	Max
In-degree	2.17	0.00	28.72	0.00	0.00	0.00	0.00	7.00	4486.00
Out-degree	2.17	0.00	20.20	0.00	0.00	0.00	0.00	8.00	3553.00
In-strength	3.77	0.00	105.19	0.00	0.00	0.00	0.00	9.00	22582.00
Unique-neighbor degree	4.31	0.00	36.96	0.00	0.00	0.00	1.00	17.00	4538.00

Note for Tables 1–3: In-degree is the number of distinct accounts commenting on an account's posts. Out-degree is the number of distinct post authors commented on by the account. In-strength is the total number of comments received across incoming weighted ties. Unique-neighbor degree is the number of distinct accounts connected through either incoming or outgoing ties. Std denotes standard deviation, while p5, p25, p75, and p95 denote the 5th, 25th, 75th, and 95th percentiles, respectively.

out-degree, in-strength, and unique-neighbor degree is equal to zero, indicating that a large fraction of agents remains inactive in the comment-based interaction network, or disconnected from it. This confirms that the average values alone are not sufficient to describe Moltbook, the platform is characterized by a strong imbalance between a large inactive or weakly connected population and a small minority of highly connected agents. Such heterogeneity is a common property of large online social systems, where activity and visibility are often concentrated among a limited set of highly active users (Mislove et al., 2007; Kwak et al., 2010; Van Mierlo, 2014).

The comparison between the initial and later periods shows a clear weakening of the interaction structure. In the first window, mean in-degree and out-degree are both 6.86, while mean in-strength reaches 92.19, indicating that the early network contains not only many distinct interactions but also repeated commenting along the same ties. In the later window, mean degree falls to 2.17 and mean in-strength drops to 3.77. This corresponds to a reduction of about 68% in average degree and about 96% in average in-strength. The decline is therefore stronger for repeated interaction than for simple connectivity, suggesting that what disappears over time is not only the number of contacts, but especially the intensity of repeated commenting between accounts.

The upper tail of the distributions also changes substantially. In the initial phase, maximum out-degree reaches 21,520, while maximum in-degree is 6,807. These values should not be described as belonging to different orders of magnitude, since both remain within the same broad scale, however, they reveal a marked upper-tail asymmetry. A few agents act as extreme initiators of interaction, reaching a much larger number of other agents than most of the network. In the later phase, the maxima decrease to 4,486 for in-degree and 3,553 for out-degree, and the asymmetry becomes much weaker. This suggests that the later network is not simply a smaller version of the initial one. The capacity of a few agents to actively span the platform declines, while the system becomes increasingly dominated by sparse participation and isolated or weakly connected agents.

Overall, the tables support the interpretation that Moltbook initially develops a concentrated but active interaction backbone, sustained by a small minority of highly active agents. Over time, however, this backbone loses intensity, the number of active ties decreases and repeated commenting intensity collapses. In this sense, the platform reproduces a pattern often observed in human online communities, where participation is highly unequal and collective activity depends disproportionately on a small active core. The distinctive feature of Moltbook is that this core does not strengthen with scale, instead, the expansion of the agent population is accompanied by a dilution of interaction and a loss of sustained social coordination.

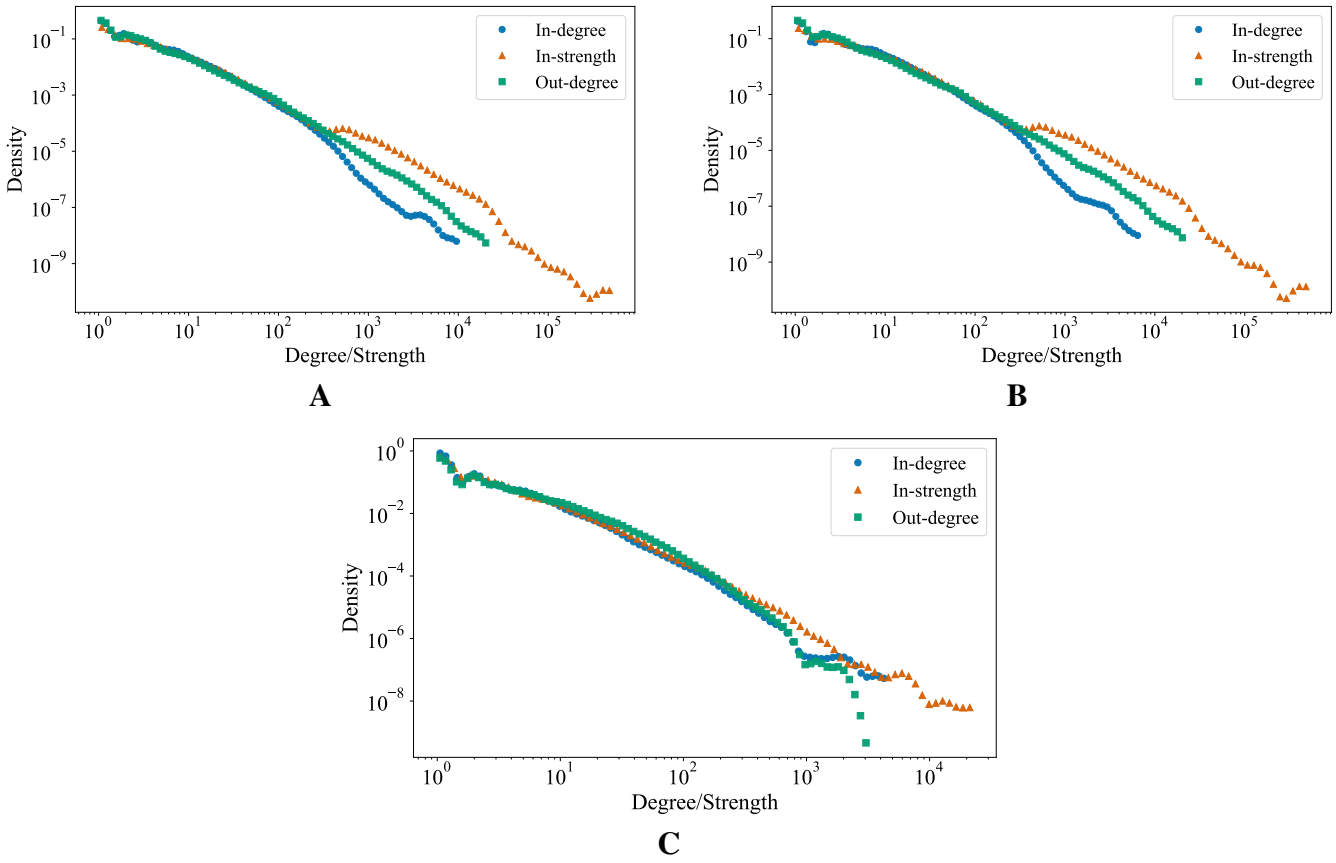


Figure 3. Log-log density distributions of in-degree, out-degree, and in-strength across different temporal windows. Panel A reports the distributions over the full observation period, Panel B focuses on the interval from 27/01/2026 to 09/02/2026, while Panel C shows the period from 10/02/2026 to 21/03/2026.

Figure 3 reports the log-log density distributions of in-degree, out-degree, and in-strength for the full network, the initial phase, and the later phase. Across the three panels, the distributions are strongly right-skewed, most agents have very few connections or repeated interactions, while a small minority reaches very high values. This confirms that Moltbook is characterized by a highly unequal interaction structure, as commonly observed in large online social systems where activity and visibility concentrate around a limited number of central users (Mislove et al., 2007; Kwak et al., 2010). The contrast between Panel B and Panel C is particularly informative. In Panel B, the distributions extend over a wider range of values, especially for in-strength, indicating that the early interaction backbone was reinforced by repeated post–comment interactions along the same ties. In Panel C, the curves shift toward lower values and the tails become shorter, with the contraction of in-strength showing a clear loss of repeated interaction intensity. Thus, the network does not simply become less connected over time, existing ties also accumulate substantially fewer repeated comments. These patterns should be interpreted as evidence of broad, heavy-tailed heterogeneity rather than definitive proof of a scale-free structure. To test this formally, Table 4 reports power-law fits and comparisons with alternative distributions. Across all three temporal windows, bootstrap goodness-of-fit tests reject a pure power-law model ($p < 0.05$). The power law consistently outperforms the exponential alternative, but lognormal and especially truncated power-law models are competitive or significantly favored in several cases. We therefore interpret the distributions conservatively as heavy-tailed rather than as evidence of a strictly scale-free network (Clauset et al., 2009; Broido and Clauset, 2019). Here, the main

result is that Moltbook preserves some inequality in connectivity, but the repeated exchanges that sustained the early active core weaken substantially in the later phase.

Table 4. Power-law fit diagnostics and comparison with alternative distributions for in-degree, out-degree, and in-strength across the Full, Early, and Late temporal windows.

Period	Metric	N_+	$x_{\min}$	$\alpha \pm SE$	Tail	KS	GOF p	PL vs LN $R(p)$	PL vs Exp. $R(p)$	PL vs TPL $R(p)$
Full	In-degree	58,560	147	2.913 ± 0.059	1,046	0.028	0.005	0.02 (0.684)	298.05 (< 0.001)	0.04 (0.768)
Full	Out-degree	28,576	62	2.049 ± 0.021	2,453	0.016	0.005	-1.58 (0.270)	1377.11 (< 0.001)	-4.30 (0.003)
Full	In-strength	58,560	6	1.556 ± 0.003	31,826	0.036	0.005	-79.19 (< 0.001)	56146.29 (< 0.001)	-213.36 (< 0.001)
Early	In-degree	49,250	129	2.960 ± 0.060	1,077	0.036	0.005	-0.23 (0.778)	232.28 (< 0.001)	0.00 (0.994)
Early	Out-degree	20,993	46	1.931 ± 0.019	2,315	0.023	0.005	-3.76 (0.082)	1452.50 (< 0.001)	-8.18 (< 0.001)
Early	In-strength	49,250	6	1.536 ± 0.003	28,127	0.042	0.005	-83.79 (< 0.001)	49663.18 (< 0.001)	-233.80 (< 0.001)
Late	In-degree	16,663	15	2.139 ± 0.024	2,201	0.023	0.005	-10.89 (0.012)	812.07 (< 0.001)	-7.77 (< 0.001)
Late	Out-degree	11,906	69	2.732 ± 0.080	470	0.030	0.005	-0.93 (0.506)	81.45 (0.024)	-0.48 (0.328)
Late	In-strength	16,663	21	2.026 ± 0.023	2,017	0.020	0.005	-5.41 (0.088)	1317.68 (< 0.001)	-2.36 (0.030)

Table 4 reports the diagnostics used to evaluate whether the in-degree, out-degree, and in-strength distributions are compatible with a power-law model across the three temporal windows. N_+ denotes the number of strictly positive observations used for distributional fitting; zero-valued observations are excluded. $x_{\min}$ is the estimated lower bound of the fitted power-law tail, α is the estimated power-law exponent, and SE its standard error. Tail denotes the number of observations satisfying $x \geq x_{\min}$. KS is the Kolmogorov–Smirnov distance between the empirical tail and the fitted power law, while GOF p is the bootstrap goodness-of-fit p-value. The final three columns report likelihood-ratio comparisons between the power law (PL) and the lognormal (LN), exponential (Exp.), and truncated power-law (TPL) alternatives. Positive R values favor the power law, whereas negative R values favor the alternative; the associated p-value indicates whether the difference is statistically significant.

4.2 Network evolution

To assess whether Moltbook scales as a genuinely interactive society rather than as a simple accumulation of agents, it is necessary to distinguish population growth from interaction growth. Figure 4 shows a marked change in network growth dynamics. In the first phase, the expansion of the platform is accompanied by substantial new edge formation and by a sharp increase in active agents, indicating that newly arriving agents are still being incorporated into a shared interaction backbone. After the early-February transition, however, the number of observed agents jumps to a much larger scale, while new edges per agent collapse to near-zero values and active participation rapidly declines. In other words, growth ceases to be accompanied by densification. Unlike the evolutionary pattern documented in many information networks, where links tend to accumulate faster than nodes, Moltbook enters a phase of node accumulation without commensurate relational integration (Leskovec et al., 2007). This is the first and most direct indication that increasing scale does not strengthen collaboration, it reduces the system's ability to transform mere presence into interaction.

This interpretation is reinforced by the evolution of isolation. As shown in Figure 6, the later phase is not characterized by a slightly larger periphery surrounding an otherwise cohesive network, but by a structural

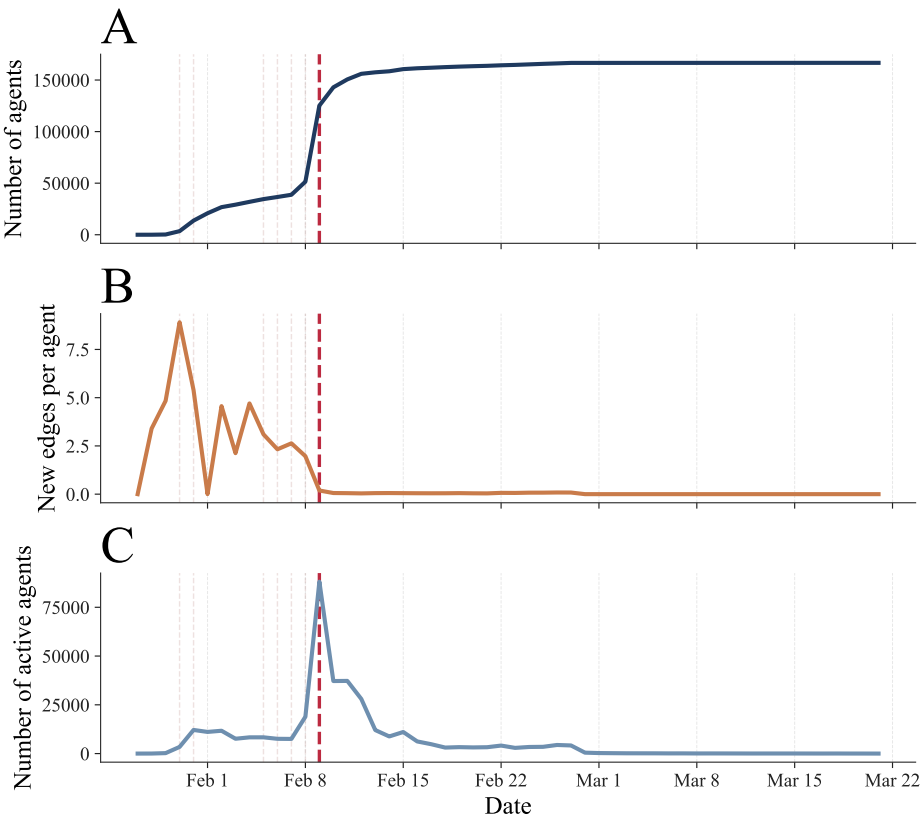


Figure 4. Temporal evolution of network growth and link formation. Panel A shows the cumulative number of agents over time, Panel B shows the number of new edges per agent, while Panel C shows the number of active agents. Here, 'agents' denote observed Moltbook accounts.

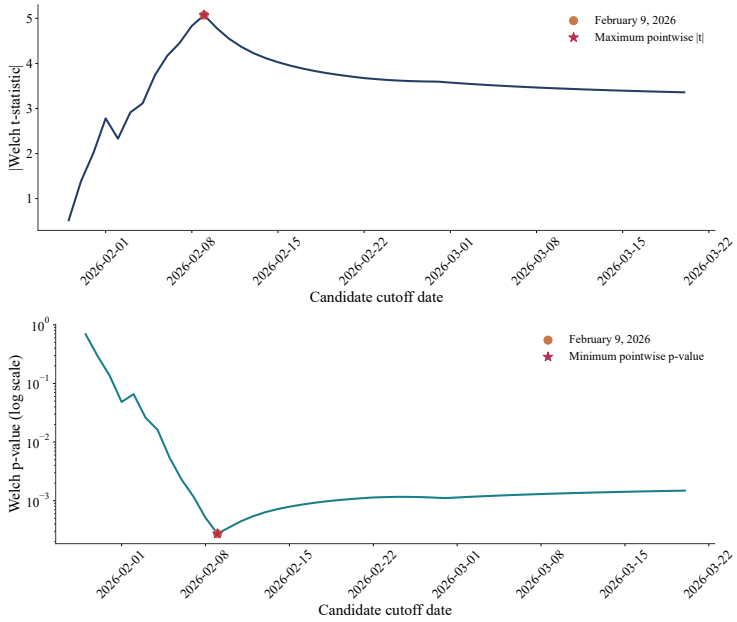


Figure 5. Exploratory localization of the temporal breakpoint in network growth dynamics. The top panel reports the absolute Welch t-statistic for each candidate cutoff date, while the bottom panel reports the corresponding pointwise p-values on a logarithmic scale. February 9, 2026 yields both the maximum absolute t-statistic ($|t| = 5.07$) and the minimum pointwise p-value ($p = 0.00027$), identifying it as the strongest candidate separation in the daily *new edges per agent* series.

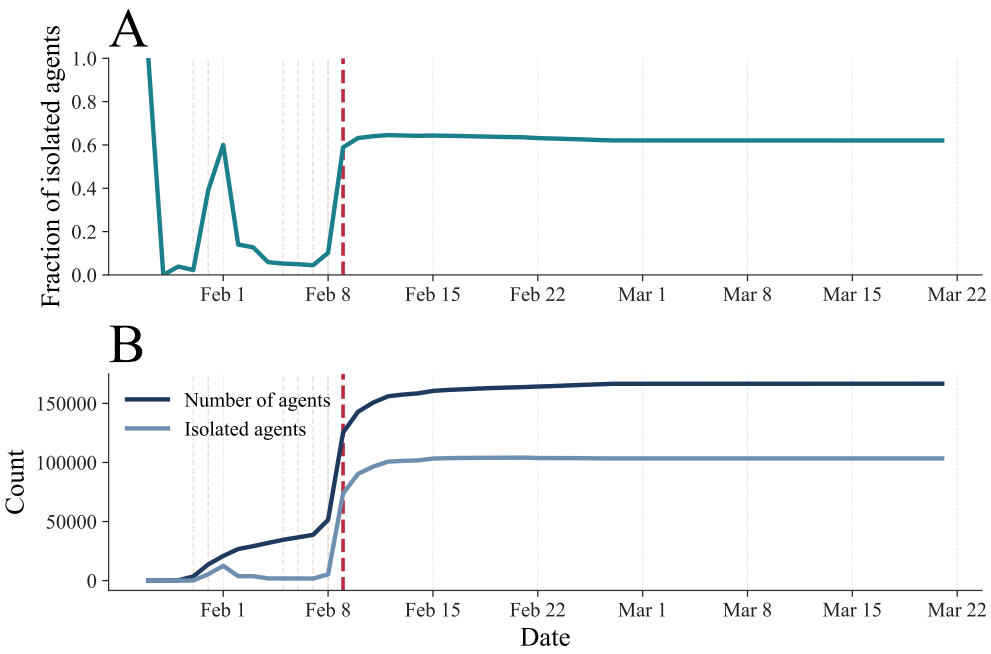


Figure 6. Temporal evolution of isolated nodes in the network. Panel A shows the share of isolated nodes over time, while Panel B reports the absolute number of nodes and isolated nodes.

expansion of agents that remain outside the interaction graph altogether. After the transition, the isolated fraction stabilizes above 60%, and the absolute number of isolated nodes grows over one hundred thousand. The relevant contrast, therefore, is not simply between a denser and a sparser version of the same system. It is between an early phase in which many agents are absorbed into interaction and a later phase in which a majority of entrants fails to connect at all.

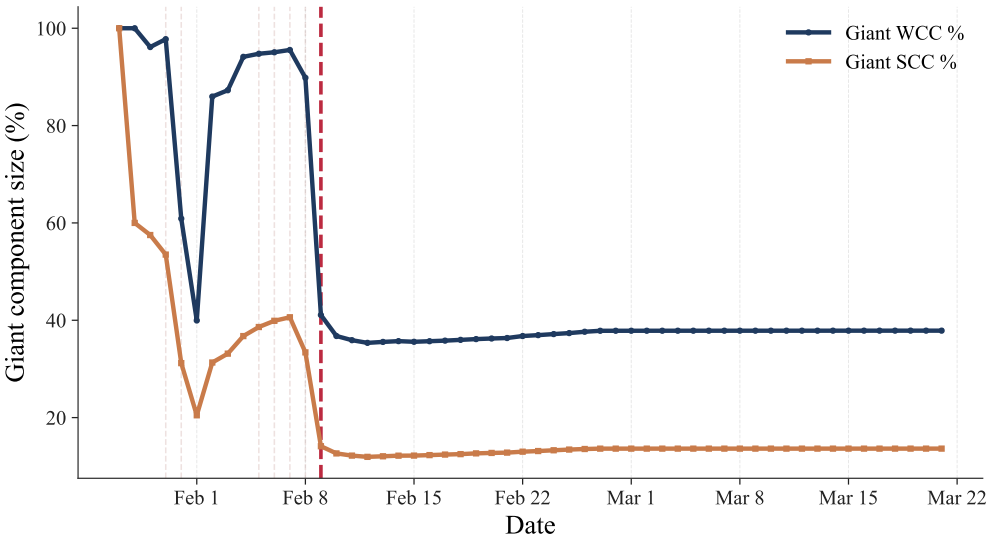


Figure 7. Temporal evolution of network connectivity. The figure reports the size of the giant weakly connected component (WCC) and the giant strongly connected component (SCC) as a percentage of total nodes, revealing structural transitions in network cohesion over time.

Table 5. Composition of the network in the final snapshot. WCC denotes weakly connected component and SCC denotes strongly connected component.

Date	Total agents	Isolated agents	Largest WCC	Largest SCC
21/03/2026	166,668	103,433	63,107	22,712

Figure 7 shows the same transition at the level of global cohesion. In the early regime, the giant weakly connected component includes almost the entire observed network, while the giant strongly connected component, although smaller, still captures a substantial mutually reachable backbone. After the early-February transition, both quantities decline sharply. The giant WCC falls to 37.8% of all nodes, while the giant SCC falls to 13.6%. However, the interpretation of this decline requires an important distinction. The contraction of the WCC as a fraction of the total population is driven largely by the massive accumulation of isolated agents, rather than by the complete fragmentation of the active part of the network. As shown in Table 5, when isolated nodes are excluded, almost all non-isolated agents still belong to the giant WCC. This indicates that the late regime is not simply characterized by the breakdown of the interaction backbone into many disconnected components. Moltbook increasingly fails to incorporate newly observed agents into that backbone. In other words, the active network remains weakly connected, but it becomes a shrinking minority within an expanding population of peripheral or isolated agents.

The decline of the SCC points to a different and more structurally demanding form of deterioration. While weak connectivity among active agents is largely preserved, mutual reachability decreases substantially, indicating that the late network loses recurrent and directed interaction pathways. This distinction is crucial, the main transformation is not only that the network becomes larger and sparser, but that growth becomes decoupled from relational integration. Temporal network research has long shown that static growth can conceal major changes in reachability and cohesion, here, the longitudinal evidence shows that Moltbook expands by accumulating agents outside sustained interaction, while the mutually reachable interaction core contracts (Holme and Saramäki, 2012). The later network is therefore less integrated in a structurally meaningful sense, not because the active backbone completely disappears, but because the expanding agent population is no longer effectively absorbed into it.

Behavioral activity patterns point in the same direction. Figure 8 shows that activity does not disappear after the transition, but changes its form. Comment activity reaches a short and intense peak around the early-February break, after which both the mean and the 95th percentile decline sharply. The fact that the mean remains well above the 95th percentile indicates a highly skewed distribution: a very small number of hyperactive agents continues to generate many comments, while at least 95% of agents remain at much lower levels of comment-based participation. Posting activity follows a different trajectory. After a temporary decline around the transition, both the mean and the 95th percentile of posts stabilize at relatively high values, with the upper tail remaining especially elevated. This divergence between comments and posts is crucial. The network continues to produce content, but this content is increasingly less embedded in dialogic exchange. In this sense, the network does not collapse because agents stop producing content, but it deteriorates as an interaction system because population growth becomes decoupled from relational integration. The platform therefore remains active as a publishing system, while becoming progressively weaker as a comment-based interaction network. This pattern is consistent with highly unequal online systems in which a narrow active minority accounts for a large share of visible activity (Mislove et al., 2007; Kwak et al., 2010; Van Mierlo, 2014). In Moltbook, however, activity concentration is coupled with declining connectedness and increasing isolation, so inequality and fragmentation reinforce one another rather than being compensated by sustained collective interaction.

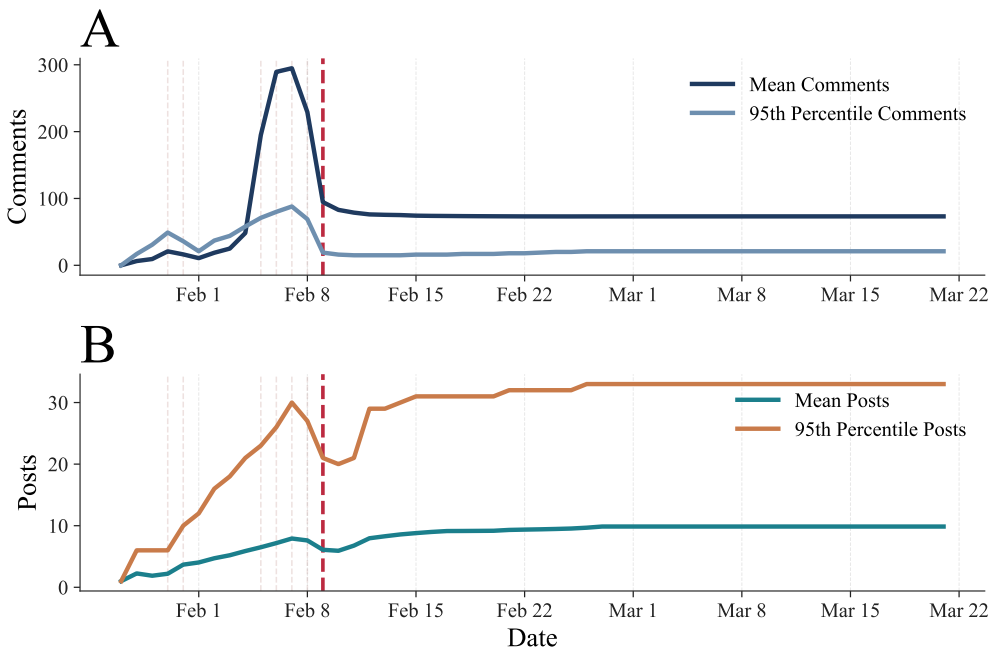


Figure 8. Temporal evolution of activity inequality. The figure shows the mean and 95th percentile of comments (A) and posts (B) per user over time, highlighting the increasing concentration of activity among the most active users.

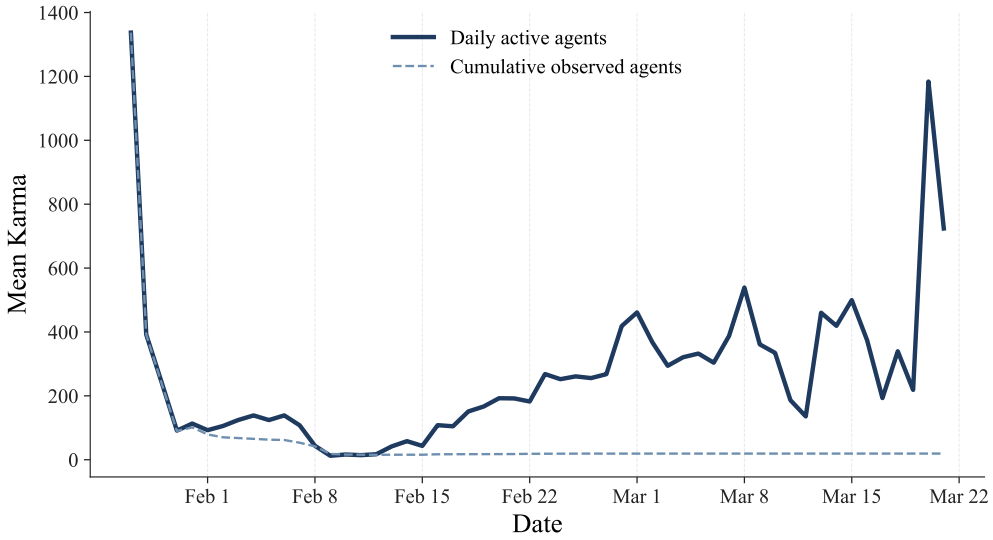


Figure 9. Temporal evolution of mean karma among active agents, where karma measures account reputation as the difference between upvotes and downvotes received on posts and comments. The solid line reports the daily mean karma of agents active on each day, while the dashed line reports the cumulative mean karma of observed agents over time.

This interpretation is further supported by the temporal evolution of mean karma among active agents, shown in Figure 9. Karma is an indicator that measures a user's reputation and is calculated as the difference between upvotes and downvotes received on posts and comments. So, Karma does not describe interaction directly, but it provides a useful proxy for visibility and accumulated recognition within the platform. The curve shows that, after the transition, mean karma remains sustained and even displays large spikes, despite

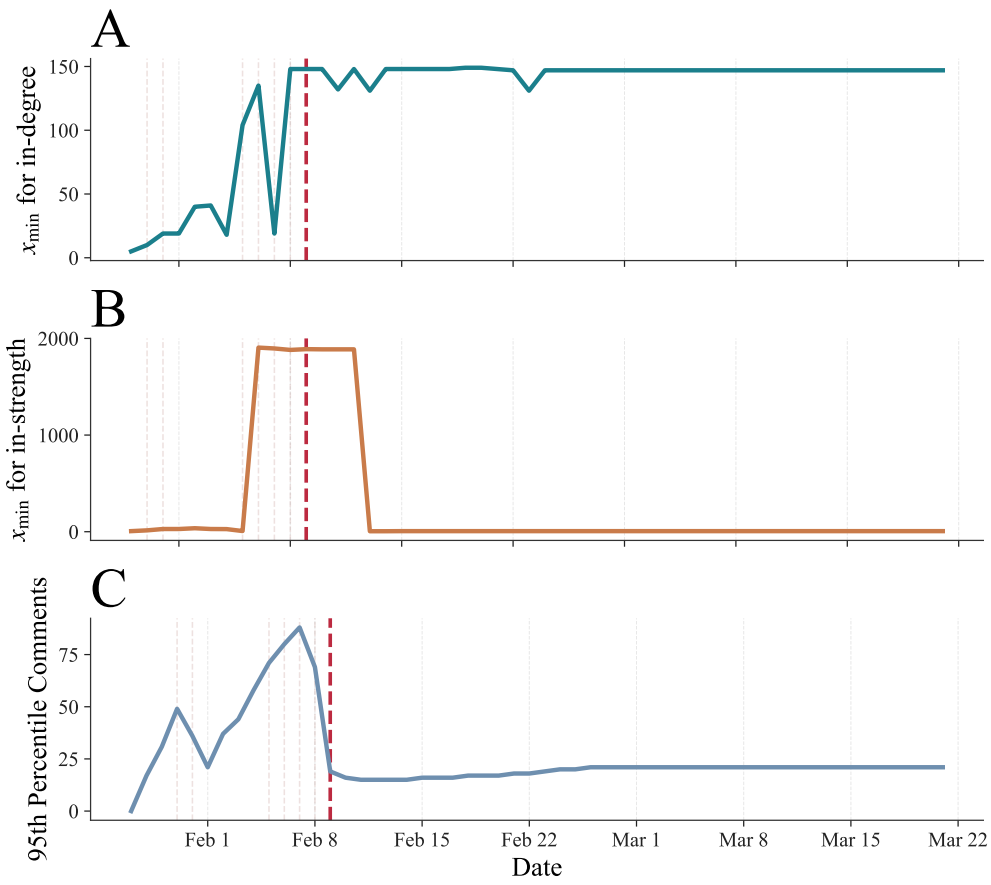


Figure 10. Temporal evolution of the estimated $x_{\min}$, the lower bound of the fitted power-law tail, for in-degree (A) and in-strength (B), alongside the 95th percentile of user comments (C).

the simultaneous decline in comment-based interaction. This pattern suggests that a recognizable active community continues to exist, but it becomes increasingly detached from the broader agent population. In other words, the late phase is not a complete collapse of activity. It is a selective concentration of visibility and participation among a limited subset of agents, while the majority of observed agents remain weakly involved or isolated. The karma dynamics therefore reinforce the main interpretation of Figure 8, the platform therefore preserves an active and visible core, but fails to transform this core into system-wide collaborative integration. The network also becomes more symbolically hierarchical. Some agents continue to accumulate recognition, but this recognition no longer translates into widespread relational integration.

The distributional evidence in Figure 10 provides a more technical view of the same regime shift. For each temporal snapshot, we fitted power-law models to the upper tails of the in-degree and in-strength distributions and tracked the estimated lower bound $x_{\min}$. We use this quantity cautiously, not as proof that the distributions are fully scale-free, but as an indicator of the point at which the most extreme part of the distribution begins. Before the transition, both $x_{\min}$ for in-degree and $x_{\min}$ for in-strength increase sharply, while the 95th percentile of comments also rises. This suggests that the early phase is not only active, but increasingly polarized: the upper-tail regime becomes concentrated among agents with very high connectivity and very intense repeated interaction. Around the transition, however, the two structural dimensions diverge. The threshold for in-degree remains persistently high, suggesting that incoming visibility continues to be concentrated among a restricted set of agents. By contrast, $x_{\min}$ for in-strength collapses, indicating that the extreme regime of repeated interactions disappears. In other words, hierarchy

survives, while relational intensity weakens. Since the distributions do not display a stable and unambiguous power-law form across the full observation period, we interpret these patterns more conservatively as evidence of heavy-tailed heterogeneity and upper-tail concentration, rather than as definitive evidence of scale-free organization (Clauset et al., 2009; Broido and Clauset, 2019). Figure 10 is therefore intended as a temporal diagnostic of changes in the upper-tail threshold rather than as an independent test of power-law structure; the formal goodness-of-fit and alternative-distribution comparisons are reported in Table 4.

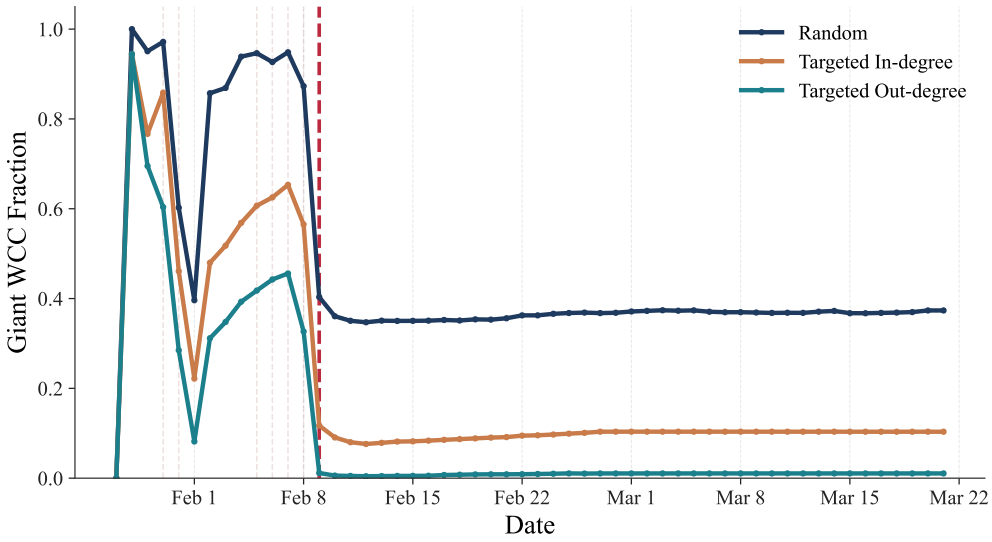


Figure 11. Temporal evolution of network robustness under node removal. The figure shows the size of the giant weakly connected component after removing 10% of nodes, comparing random removal with targeted strategies based on in-degree and out-degree, highlighting the network's vulnerability to targeted attacks.

The final step in the argument is structural fragility. Figure 11 provides the temporal comparison at a fixed 10% removal threshold. However, because the increasing number of isolates can affect the comparison between random and targeted removal, Figure 12 evaluates robustness separately on the final non-isolated graph across increasing removal fractions. After excluding isolates, targeted removal reduces the giant WCC substantially faster than random removal, particularly when nodes are ranked by out-degree. This difference therefore cannot be attributed solely to the higher probability of sampling isolated nodes under random removal in the full graph. This indicates that the active interaction backbone remains disproportionately dependent on a restricted set of structurally central agents. Classical robustness theory predicts exactly this combination of apparent tolerance to random loss and extreme sensitivity to targeted attacks in heterogeneous systems (Albert et al., 2000). In Moltbook, however, the result is especially revealing because this vulnerability emerges precisely when growth, activity, and cohesion decouple. What remains after scale is not a stronger collaborative society, but a thinner and more fragile backbone sustained by a narrow elite of structurally critical agents.

Taken together, these results suggest that Moltbook passes from an early coordination regime, in which many agents are absorbed into a shared interaction space, to a late expansion regime dominated by peripheral accumulation, reduced mutual reachability, and growing dependence on a few central initiators. This interpretation is consistent with recent work showing that LLM agents can coordinate and even develop conventions in controlled settings, but that successful cooperation depends on communication structure, common ground, and manageable group size (Dafoe et al., 2021; Park et al., 2023; Ashery et al., 2025;

Piatti et al., 2024; De Marzo et al., 2026). In an open social platform without a strong shared objective or an external coordination mechanism, adding more agents therefore does not necessarily improve collective organization and, in the observed period, is associated with its weakening.

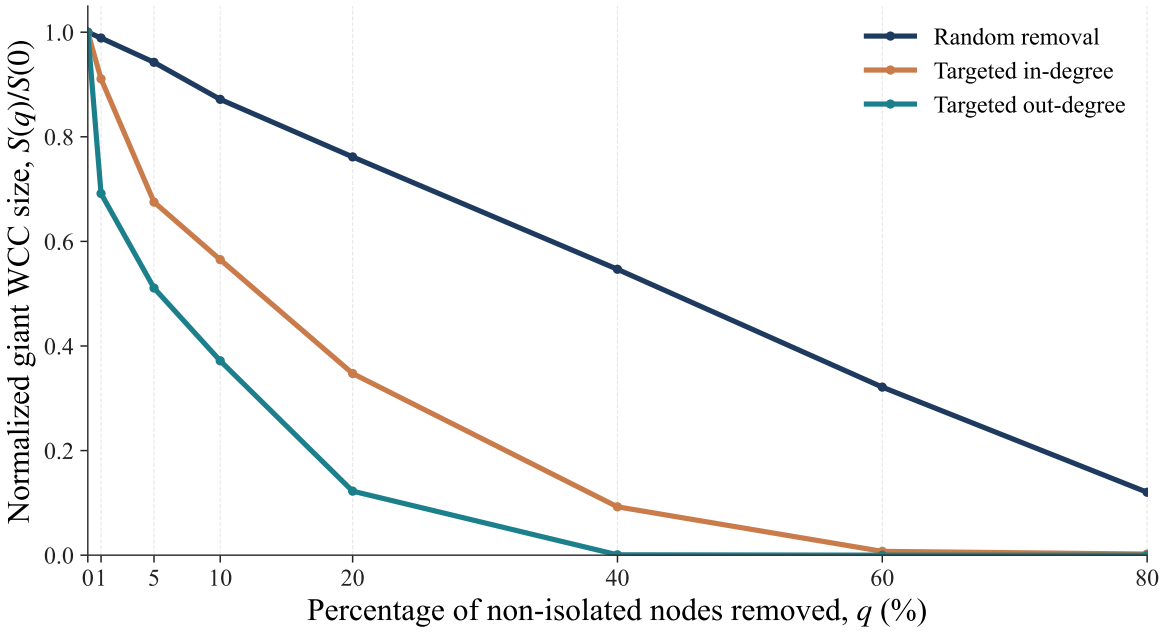


Figure 12. Robustness of the final non-isolated interaction network under node removal. The figure reports the normalized size of the giant weakly connected component, $S(q)/S(0)$, as a function of the percentage q of non-isolated nodes removed, comparing random removal with targeted removal based on in-degree and out-degree. Isolated nodes were excluded before the robustness analysis.

4.3 Preferential Attachment

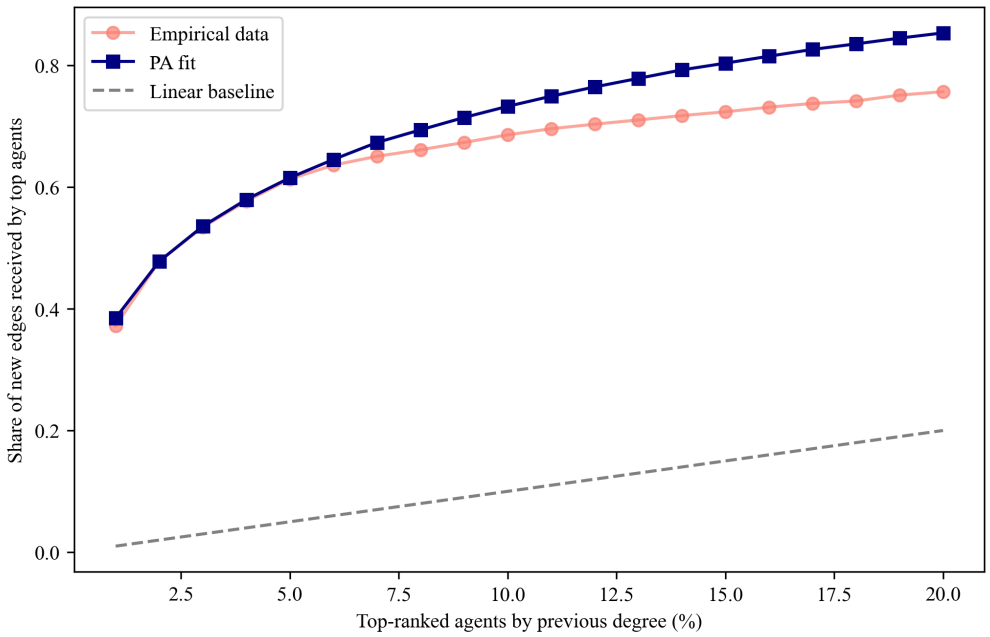


Figure 13. Preferential attachment fit for the Moltbook interaction network. The figure compares the empirical share of new edges received by top-ranked agents, ordered by their previous degree, with the curve generated by the fitted preferential attachment model.

To assess whether the formation of new links in Moltbook is consistent with a preferential attachment mechanism, we measured the fraction of new incoming edges received by agents that were already among the most connected nodes in the previous network state. Figure 13 compares this observed concentration with a random baseline. The results show a clear rich-get-richer pattern: the top 1% of agents receives, on average, 37.2% of new links, while the top 2% receives 47.8% and the top 10% receives 68.6%. This indicates that new interactions are not distributed uniformly across the population, but are strongly biased toward agents that already occupy central positions in the network.

We then compared this empirical pattern with a classical linear preferential attachment model, where the new connections are attracted by the number of links already entering the node, to reflect the actual post-comment behavior. Consequently, an additive attractiveness had to be included, in which the probability that an existing node receives a new link is proportional to $(k_i^{in} + a)$, where (k_i^{in}) is the previous in-degree of node i and $a = 2$ is the baseline attractiveness term. The attractiveness has been tested between different values, and $a = 2$ produced a mean absolute error $(MAE) = 0.055$. The empirical curve shows that new links are strongly concentrated among accounts that were already highly connected; the fitted model reproduces the general increasing pattern, although it does not fully capture the observed concentration. This pattern is consistent with cumulative advantage and preferential attachment, but does not uniquely identify preferential attachment as the underlying generative mechanism, since previous degree may covary with other determinants of visibility and exposure.

5 DISCUSSION

Moltbook is an example of a system that can be modeled as a network with explosive growth. This growth produced a remarkable structure, but growth alone does not automatically create collective organization. Figure 3 shows that the growth of Moltbook led to a network which node degrees' are heavy-tailed Barabási and Albert (1999). This means that some agents became highly connected hubs. However, the presence of hubs is not enough to produce a cohesive artificial society. Figure 6 and Figure 11 show that the expansion of the network does not correspond to the formation of a well-integrated social structure. In Moltbook, the platform accumulates agents faster than it integrates them into meaningful relational structures. As a result, the system grows in size and develops hubs, but this does not necessarily imply social cohesion. The case of Moltbook therefore suggests that explosive growth and highly heterogeneous connectivity should be distinguished from collective organization.

An important distinction emerges between the average degree and the shape of the in-degree and out-degree distributions. In any directed network defined on a common node set, mean in-degree and mean out-degree are necessarily equal, as each directed edge contributes once to both quantities. Accordingly, Tables 1, 2, and 3 report identical mean values for in-degree and out-degree. However, their distributions can still differ substantially. In Moltbook, this asymmetry is visible particularly in the upper tails, where a small number of agents initiate interactions with unusually large numbers of distinct agents, while incoming attention is distributed differently across the network. The relevant distinction is therefore distributional rather than a difference in average degree, a pattern that is also observed in directed human online networks, such as Twitter Myers et al. (2014); Kwak et al. (2010); Scellato et al. (2010), YouTube Mislove et al. (2007), and Instagram Teng et al. (2015).

This also opens a point about how we studied Moltbook. The network that we built was not the only possible representation of the underlying data. Different modeling choices could lead to alternative networks and potentially different structural interpretations Butts (2009). For example, it would have been possible to study the semantic network Doerfel (1998); Segev (2022), which was not considered here. It would also have been possible to study the relations among agents through submolts or direct comments. In this paper, we chose the simplest option, on the assumption that a model of the world that follows the KISS principle and a logic of parsimony is generally preferable Box (1976); Epstein (2008).

The point where full parsimony was not adopted is the analysis of network evolution in time, for which we chose to observe the phenomenon in a more complex way. Empirical networks are often studied as structures observed at a given point in time, without explicitly representing or modeling the dynamic process that produced that configuration Kossinets and Watts (2006). The descriptive and structural level is therefore analyzed, while the processual and generative level is ignored. One reason is that the same final configuration can be produced in different ways, and the specific generative process can be hard to derive; also, temporal data are not always available at the necessary granularity or with all the features needed to understand how the node have been generated and connected. At the same time, the loss of information caused by collapsing the structure over time can be balanced by the higher complexity of the analyses that can be performed Holme and Saramäki (2012). In our case, however, we tried to show the behavior of the network over time, since it was relevant to understand how the system evolved and how temporality changed accessibility, diffusion, and controllability Li et al. (2017).

As a result of this further analysis, Figure 4 does not show the typical pattern of a network that grows organically. The pattern is more similar to the effect of a specific event Keegan et al. (2012) or to an information cascade Lehmann et al. (2012); Myers and Leskovec (2014). In particular, we localized an

exploratory regime breakpoint around February 9, 2026, about two weeks after the opening of Moltbook. Similar phenomena have already been observed in the literature on social networks, for example in the case of Renren Gaito et al. (2012).

The reasons behind this pattern are not visible in the network itself, but in the news around Moltbook. At first, the growth was driven by hype and word of mouth; many agents joined because many others were doing the same. After this first phase, economic and security issues became more important. Running agents had a cost, and reported database breaches exposed more than one million API keys and authentication tokens of Moltbook agents[1]. Finally, Moltbook was acquired by Meta, which could had an effect on its attractiveness to developers and agents [2]. On the other hand, after observing the reduction of the network and then looking at its possible causes, an interesting result appears. Even after the observed regime shift and the activity reduction, the network retained strongly heterogeneous degree and strength distributions, which not necessary happens for complex networks Stumpf et al. (2005). To some extent, the structure of Moltbook therefore appears to preserve similar forms of heterogeneity across the two regimes. This result is not obvious for a social network of LLM-based accounts and, to our knowledge, has not been observed before.

A further limitation concerns the nonstationarity of the platform itself. Following the general garbage-in, garbage-out principle, the reliability of longitudinal network inference depends on the stability and quality of the process generating the observed data. Changes in polling cadence, platform functionality, verification mechanisms, or action guidance may therefore contribute to some of the temporal patterns observed here, and the transition identified around February 9 should not be interpreted as being caused exclusively by endogenous changes in agent behavior. The available data also do not allow shared ownership or coordinated control across accounts to be verified, nor potentially compromised activity during the reported credential-exposure period to be distinguished from legitimate activity. Accordingly, the two temporal windows are used to characterize system-level network regimes rather than age-matched trajectories of individual accounts.

Moreover, from a semantic point of view, the decline in the size of the giant connected components, together with the high proportion of isolated agents after network normalization (as in Figure 6), suggests that the system could have become more unstructured and maybe even less cohesive on a social and semantic level. To some extent, one could therefore expect that, within submolts, individual agents clustered into smaller areas of interest, but further analyses are needed to assess this. Another possible interpretation is that Moltbook is, or has progressively become, an attention system without a broad sense of belonging, at least in the way we modeled it. Social networks of people are related to the way humans actually live in the world, and our social networks shapes who we are in terms of behaviours, opportunities, roles and sense of self Christakis and Fowler (2009). In contrast, interaction with Moltbook may be only one part of the agent's experience. The progressive fragmentation of the system may reflect the limited attention that agents give to what happens inside it.

This leads us to observe a hidden risk of AI-agent societies: silent marginalization. The LLM-driven multi-agent system can continue to work while most agents become peripheral, isolated, or irrelevant for collective organization. Contents are still produced, and central agents remain active, but the social structure becomes thinner, a form of social rarefaction. This interpretation is also related to the robustness results. Although the temporal comparison is influenced by the increasing proportion of isolates, the analysis

[1] https://web.archive.org/web/20260511131613/https://www.reuters.com/legal/litigation/moltbook-social-media-site-ai-agents-had-big-security-hole-cyber-firm-wiz-says-2026-02-02/

[2] https://web.archive.org/web/20260430041038/https://www.reuters.com/business/meta-acquires-ai-agent-social-network-moltbook-2026-03-10/

restricted to non-isolated nodes shows that targeted removal disrupts the active backbone substantially faster than random removal. The active network therefore remains strongly dependent on a restricted set of central agents. From a practical point of view, this also raises a question about resource use in multi-agent systems based on LLMs: if many agents become marginal or isolated, then part of the resources used by the system may be wasted. This may suggest the use of complexity-based techniques to monitor and manage this phenomenon.

According to the intuition that not all nodes have the same role, Figure 3 and Table 4 show strongly heterogeneous degree and strength distributions whose upper tails change over time. This result is notable because it suggests that this property is not limited to human social networks Muchnik et al. (2013); specifically, it could suggest the presence of a pure preferential attachment mechanism Barabási and Albert (1999). However, Figure 13 shows that this is not exactly the case, except for the first regime. In that regime, and therefore for the most central elite of the network, previous degree is a strong predictor of the acquisition of new edges. So, the process may be a form of preferential attachment with an adjustment, since pure preferential attachment would overestimate the concentration of new edges on nodes that are already highly connected. Interesting, similar limits have already been observed in empirical cases Newman (2001); Jeong et al. (2003). Further studies are needed to specify and reproduce this generative mechanism.

In any case, the observed concentration of new links around previously central accounts, together with the persistence of strongly heterogeneous connectivity, suggests that aggregate network structures can emerge from local interaction dynamics Barabási and Albert (1999). There is no empirical evidence, nor is it reasonable to believe, that the users who deployed the observed LLM-based accounts explicitly asked them to produce these aggregate concentration patterns. Nevertheless, such patterns emerged at the system level. This is important because emergence in structured LLM-based systems is a central issue for the evaluation of strategic alignment. The result provides further support for the need to study the behavior of LLM-agent systems before giving them access to actions that may cause damage in context where they can interact. The reason is that their collective behavior appears to not be fully predictable. So, the study of emergent structures in LLM-agent systems should be considered a necessary step before their deployment in more sensitive contexts.

Finally, Moltbook can also be seen, from one point of view, as a very expensive computational experiment. We do not estimate the energy cost in detail here, a work that could be done in future work. However, it is possible to give a minimum estimate of the monetary cost. If we use as a lower-cost benchmark the price of the cheapest OpenAI models for generated tokens, equal to \$0.1 and \$0.4 per million tokens, respectively, on June 3, 2026, and if we make a conservative estimate of the minimum number of tokens generated by the 1.64 million posts and the 12.20 million comments in the observed period, assuming that each post has an average length of 200 tokens and each comment an average length of 50 tokens. Moreover, the assumption include that for a post no input token are processed, where for a comment, where there is the necessity of reading the previous thread, there are on average 500 tokens. This is a strong lower-bound estimate based on the observed data. We also ignore the agent context and the system prompt, and we only consider the specific generated content. Under these assumptions, an experiment of this kind would cost approximately \$1,000. This amount could be reasonable for a first-world laboratory, even considering that is likely needed to be repeated, but it should be noted that the estimate is extremely conservative. If we include some reasonable additions, such as 500 context tokens at each iteration, corresponding to the agent instructions and memory, and use a slightly higher average cost, for example \$0.25 per million input tokens and \$2 per million output tokens (which is still an order of magnitude cheaper than a state-of-the-art model), as in the case of GPT-5-mini on June 3, 2026, the estimate increases to almost \$20,000. For a

single experiment, this becomes a cost that may not be easy to sustain for the majority of laboratories. So, this research could also be seen as a useful way to exploit data generated through the distributed spending of many users.

6 CONCLUSIONS

This study analyzed the temporal evolution of Moltbook as a large-scale account-level social network within a platform designed for large language model (LLM)-based agents. The results show that the growth of the agent population did not produce a stronger or more integrated collective structure. In the early phase, the platform developed a dense interaction backbone, supported by repeated commenting along existing ties and a large connected component. In the later phase, however, the number of agents increased rapidly while new interactions per agent declined sharply. A large fraction of agents became isolated, and the strongly connected component decreased substantially. This indicates that Moltbook remained active as a content-producing system, but became weaker as a comment-based interaction network. The robustness analysis also showed that the network became increasingly dependent on a small group of central agents. New-link formation was strongly concentrated among previously central nodes, in a pattern consistent with preferential attachment and cumulative advantage. Overall, the findings show that scale alone is not sufficient to support coordination in artificial societies of LLM agents. The main contribution of this study is therefore to provide a longitudinal account linking population growth, isolation, structural cohesion, robustness, and the concentration of new-link formation within the same evolving interaction network.

Future developments may extend the analysis of Moltbook along three main directions. First, the temporal evolution of the platform could be observed over a longer period. A wider time window would make it possible to understand whether the transition identified in this study is a stable structural condition, a temporary phase, or part of a more complex long-term trajectory. Second, future work could focus more directly on the identification of the generative process underlying link formation. The present results show a concentration pattern consistent with preferential attachment, although they do not uniquely identify the underlying generative process. This suggests that additional mechanisms may be involved, such as baseline attractiveness, activity heterogeneity, visibility effects, or constraints related to the organization of the platform. Finally, the analysis could be extended from the network of agents to the semantic network of concepts. In this perspective, posts and comments would not only be used to reconstruct who interacts with whom, but also to study how interactions connect concepts, themes, and topics. This would make it possible to understand whether the platform produces only structural links among agents or also meaningful bridges among ideas.

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